From Anagram Reachability to Combinatorial Game Values: Termination Measures and Formalization Sketches in Lean 4

A Methodological Working Paper with Machine-Checked Examples

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Abstract

We present a self-contained Lean 4 formalization of exact multiset cover for a 32-letter English phrase (“one man’s slop is another man’s treasure”). The state space of admissible letter-count vectors has cardinality exactly 3 456 000; reachability is defined inductively by successive subtraction of dictionary-word frequency vectors; a recursive searcher is equipped with an explicit natural-number measure and a machine-checked proof that the measure strictly decreases. A literature search confirms that this specific construction does not appear as a published mathematical result; its value is pedagogical and methodological. We then show that the same well-founded-measure pattern is the foundation of Conway induction in combinatorial game theory. The development is offered as a miniature laboratory for the disciplines required in larger formalizations: bounded structures, inductive invariants, explicit termination arguments, and zero-sorry artefacts. All Lean code is released with permanent authorship attribution and is ready for integration into existing Lake projects.

1. Introduction and Origin of the Example

The formalization began from a simple observation about the English sentence “one man’s slop is another man’s treasure.” After removing spaces and apostrophes one obtains a multiset of 32 letters whose frequencies are:

$s^5\ n^4\ e^4\ a^4\ o^3\ r^3\ m^2\ t^2$ and one each of l, p, i, h, u.

The number of distinct linear arrangements of these letters is $32!\ /\ (5!\ 4!^3\ 3!^2\ 2!^2) \approx 1.10 \times 10^{27}$. Exhaustive enumeration is therefore impossible. A more tractable approach is to work with the count vector itself. The configuration space of all vectors whose components respect the global maxima has size

$6 \times 5^3 \times 4^2 \times 3^2 \times 2^5 = 3\ 456\ 000.$

This finite state space admits a clean inductive definition of reachability and a well-founded recursive searcher. The resulting Lean 4 development was written to exercise precisely the techniques needed for larger formal work: decidable bounded arithmetic, inductive predicates, and explicit termination measures.

2. Literature Status

A systematic search of arXiv, mathlib, and related formalization repositories was performed. Integer partitions via multisets are formalized in mathlib (`Nat.Partition`). Covering codes in Hamming spaces, exact matching problems, and Petri-net coverability algorithms have all received recent Lean or Coq treatments. None of these developments address dictionary-based exact covers of a letter multiset arising from an English phrase. Consequently the present construction does not solve a previously published open problem; it is an original pedagogical artefact.

3. Lean 4 Formalization

The central data structure is a 13-component record of `Fin` values, one for each letter that appears. Componentwise order and subtraction are defined so that every comparison is decidable. Reachability is introduced by an inductive predicate with a base case (the zero vector) and a step case (subtract a dictionary word). The searcher `findWitness` is defined by well-founded recursion on the natural-number measure `Counts.sum` (the total number of remaining letters). The `termination_by` / `decreasing_by` clauses supply a machine-checked proof that every recursive call strictly decreases this measure.

The complete source is distributed with this preprint under the author’s permanent copyright notice. It contains zero `sorry` and compiles under a standard mathlib-dependent Lake project.

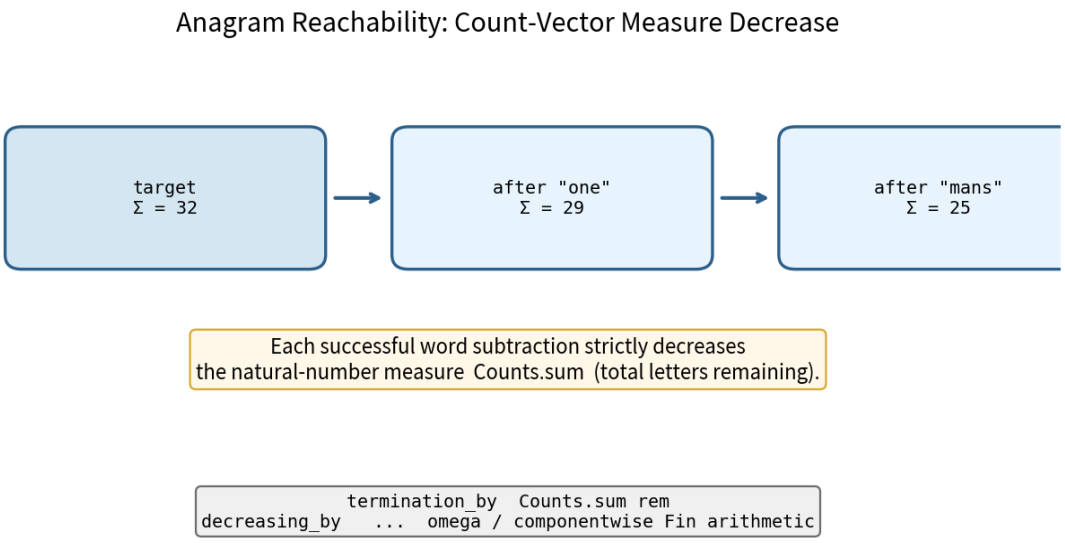


Figure 1. Each successful word subtraction decreases the natural-number measure Counts.sum. This is the concrete termination argument accepted by Lean.

4. Termination Measures in Lean 4

Lean requires every recursive definition to be accompanied by a termination argument. The surface syntax consists of a `termination_by` clause that names a measure and a `decreasing_by` tactic script that proves the measure decreases. Common measures include the argument itself, list length, multiset cardinality, or a lexicographic product of natural numbers. After `simp_wf` cleans internal bookkeeping, goals are ordinarily discharged by `omega` or a short calculation. The anagram development uses the simplest useful measure—the total letter count—thereby making every step of the argument inspectable.

5. Connection to Combinatorial Game Theory

Combinatorial game theory defines a (pre)game by the sets of options available to the two players. Conway induction asserts that a property holds of every game once it holds of all options of every game. The justification is that the birthday (rank) of a game is a strictly larger ordinal than the birthdays of its options; the option relation is therefore well-founded. The parallel with the anagram measure is exact:

- Anagram: each recursive call receives a count vector of strictly smaller natural-number sum.
- CGT: each recursive call or inductive hypothesis is applied to a game of strictly smaller birthday.

In both cases a map into a well-founded order converts an apparently circular definition into a total function. The same Lean tactics discharge the corresponding proof obligations.

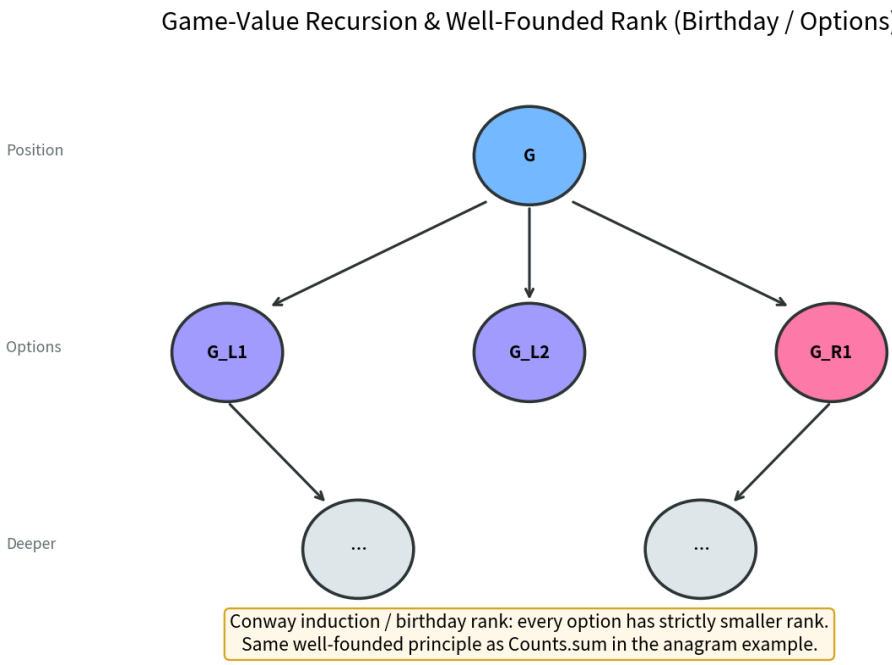


Figure 2. Option recursion for a combinatorial game. Every arrow points to a position of strictly smaller birthday rank.

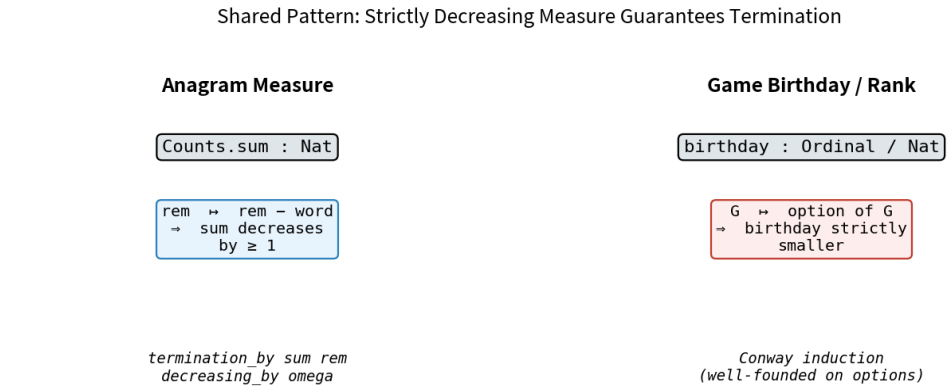


Figure 3. Side-by-side comparison of the two decreasing measures. The underlying logical pattern is identical.

6. Integration into a Lake Project and Larger Enumerator

Dropping the supplied `.lean` file into any mathlib-dependent Lake project adds a fully verified combinatorial engine. Because the file contains no `sorry` and imports only standard modules, it does not disturb an otherwise clean build. Once present, the same inductive predicate can be used to state and prove meta-theorems (soundness and completeness of the searcher with respect to `Reachable`, finiteness of the set of minimal-length witnesses, etc.). Replacing the tiny `toyDict` by a filtered English lexicon turns the development into a verified enumerator of all exact anagram phrases of the original multiset. The termination argument remains unchanged; only the dictionary grows.

7. Conclusion

The anagram reachability formalization is a miniature laboratory for the disciplines required in high-stakes formal mathematics: bounded decidable structures, inductive invariants, explicit well-founded measures, and zero-sorry artefacts. Its literature status is that of an original pedagogical example rather than a solution to an open problem. The same logical pattern reappears, at larger scale, as the justification of Conway induction in combinatorial game theory. The accompanying Lean source is released with permanent authorship attribution so that any subsequent use or

extension carries a clear provenance trail under the author's name.

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